

Applied Optimization

Solution to Exercise 3 - Convex Optimization Problems

QCQP to SOCP (0.5 pt)

Convert the following QCQP into SOCP:

$$\begin{aligned} \text{minimize} \quad & x_1^2 + 4x_1x_2 + 4x_2^2 \\ \text{subject to} \quad & 9x_1^2 + 16x_2^2 \leq 25 \\ & x_1 - x_2 = 1 \end{aligned}$$

Solution: The QCQP can be written in Epigraph form :

$$\begin{aligned} \text{minimize} \quad & t \\ \text{subject to} \quad & x_1^2 + 4x_1x_2 + 4x_2^2 \leq t \\ & 9x_1^2 + 16x_2^2 \leq 25 \\ & x_1 - x_2 = 1 \end{aligned}$$

Consider a quadratic constraint of the form

$$x^T A^T A x + b^T x + c \leq 0.$$

It is equivalent to the SOC constraint

$$\left\| \begin{pmatrix} 1 + (b^T x + c)/2 \\ Ax \end{pmatrix} \right\|_2 \leq (1 - (b^T x + c))/2$$

Here is the proof of equivalence. We can transform it by adding additional terms

$$\begin{aligned} x^T A^T A x + (b^T x + c)/2 + (b^T x + c)^2/4 + 1/4 &\leq (b^T x + c)^2/4 - (b^T x + c)/2 + 1/4. \\ \Rightarrow (Ax)^2 + ((1 + (b^T x + c))/2)^2 &\leq ((1 - (b^T x + c))/2)^2 \end{aligned}$$

If $1 - (b^T x + c) \geq 0$, By taking square root of both sides, we can get

$$\left\| \begin{pmatrix} 1 + (b^T x + c)/2 \\ Ax \end{pmatrix} \right\|_2 \leq (1 - (b^T x + c))/2$$

If $1 - (b^T x + c) < 0 \Rightarrow b^T x + c > 1 \Rightarrow ((1 + (b^T x + c))/2)^2 > ((1 - (b^T x + c))/2)^2$, and because $(Ax)^2 \geq 0$, the quadratic constraint has no solution. In the SOC, since the norm is always larger than or equal to 0, the inequality doesn't hold. The SOC also has no solution.

In summary, the QC is equivalent to the SOC constraint.

In the specific problem, the first constraint is $x^T \begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & 0 \end{pmatrix} x + \begin{pmatrix} 0 & 0 & -1 \end{pmatrix} x \leq 0$,

where $x = (x_1, x_2, t)^T$. We can factorize out the matrix $A^T A$ and get $A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

Hence the first SOC constraint is

$$\left\| \begin{pmatrix} (1-t)/2 \\ 1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} x \right\|_2 \leq (1+t)/2 .$$

Then second constraint is simply

$$\left\| \begin{pmatrix} 3x_1 \\ 4x_2 \end{pmatrix} \right\|_2 \leq 5$$

So, simplifying the expressions and rewriting them in the matrix form, the final SOCP is:

$$\begin{aligned} & \text{minimize } t \\ & \text{subject to } \left\| \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & -0.5 \end{pmatrix} x + \begin{pmatrix} 0 \\ 0.5 \end{pmatrix} \right\|_2 \leq \begin{pmatrix} 0 & 0 & 0.5 \end{pmatrix} x + 0.5 \\ & \left\| \begin{pmatrix} 3 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{pmatrix} x \right\|_2 \leq 5 \\ & \begin{pmatrix} 1 & -1 & 0 \end{pmatrix} x = 1 \end{aligned}$$

Linear Programming

Transform (0.5 pt)

For the following optimization problem

$$\begin{aligned} & \text{minimize } \|(2x + 3y, -3x)^T\|_\infty \\ & \text{subject to } |x - 2y| \leq 3 \end{aligned}$$

(1) Express the problem as a linear program. (2) Convert the LP so that all variables are in $\mathbb{R}_+$ and there is no other inequality constraints that **each individual variable** is ≥ 0 .

Solution:

(a) The linear program is as follows:

$$\begin{aligned}
& \text{minimize} && t \\
& \text{subject to} && t \geq 2x + 3y \\
& && t \geq -(2x + 3y) \\
& && t \geq -3x \\
& && t \geq -(-3x) \\
& && x - 2y \leq 3 \\
& && -(x - 2y) \leq 3
\end{aligned}$$

(b) By introducing slack variables, we can eliminate inequality constraints.

$$\begin{aligned}
& \text{minimize} && t \\
& \text{subject to} && -t + 2x + 3y + s_1 = 0 \\
& && -t - 2x - 3y + s_2 = 0 \\
& && -t - 3x + s_3 = 0 \\
& && -t + 3x + s_4 = 0 \\
& && x - 2y - 3 + s_5 = 0 \\
& && -x + 2y - 3 + s_6 = 0 \\
& && s_1, s_2, s_3, s_4, s_5, s_6 \geq 0
\end{aligned}$$

Furthermore, to make sure all variables are in $\mathbb{R}_+$, we need to add more slack variables. Since the original objective function is always ≥ 0 because of the infinity norm, $t \geq 0$. The problem is converted to:

$$\begin{aligned}
& \text{minimize} && t \\
& \text{subject to} && -t + 2(x^+ - x^-) + 3(y^+ - y^-) + s_1 = 0 \\
& && -t - 2(x^+ - x^-) - 3(y^+ - y^-) + s_2 = 0 \\
& && -t - 3(x^+ - x^-) + s_3 = 0 \\
& && -t + 3(x^+ - x^-) + s_4 = 0 \\
& && (x^+ - x^-) - 2(y^+ - y^-) - 3 + s_5 = 0 \\
& && -(x^+ - x^-) + 2(y^+ - y^-) - 3 + s_6 = 0 \\
& && t, x^+, x^-, y^+, y^-, s_1, s_2, s_3, s_4, s_5, s_6 \geq 0
\end{aligned}$$

Transform general LP to standard form (Bonus 0.5 pt)

A general linear program has the form

$$\begin{aligned}
& \text{minimize} && c^T x + d \\
& \text{subject to} && Gx \preceq h \\
& && Ax = b
\end{aligned}$$

where $G \in \mathbb{R}^{m \times n}$ and $A \in \mathbb{R}^{p \times n}$.

Transform the general LP to its standard form:

$$\begin{aligned} & \text{minimize} && p^T x' \\ & \text{subject to} && Bx' = e \\ & && x' \succeq 0 \end{aligned}$$

Explain in detail the relation between the feasible sets, the optimal solutions, and the optimal values of the standard form LP and the original LP.

Solution: The first step is to introduce slack variables s_i for the inequalities, which results in

$$\begin{aligned} & \text{minimize} && c^T x + d \\ & \text{subject to} && Gx + s = h \\ & && Ax = b \\ & && s \succeq 0. \end{aligned}$$

The second step is to express the variable x as the difference of two nonnegative variables x^+ and x^- , i.e., $x = x^+ - x^-$, $x^+, x^- \geq 0$. This yields the problem

$$\begin{aligned} & \text{minimize} && c^T x^+ - c^T x^- + d \\ & \text{subject to} && Gx^+ - Gx^- + s = h \\ & && Ax^+ - Ax^- = b \\ & && x^+ \succeq 0, x^- \succeq 0, s \succeq 0. \end{aligned}$$

To remove the constant d in the objective, we can introduce two variables d^+, d^- . The problem is

$$\begin{aligned} & \text{minimize} && c^T x^+ - c^T x^- + d^+ - d^- \\ & \text{subject to} && Gx^+ - Gx^- + s = h \\ & && Ax^+ - Ax^- = b \\ & && d^+ - d^- = d \\ & && x^+ \succeq 0, x^- \succeq 0, s \succeq 0, d^+ > 0, d^- > 0. \end{aligned}$$

Which is the standard form equivalent of the original LP.

Suppose x is feasible in the original LP. Define $x_i^+ = \max\{0, x_i\}$, $x_i^- = \max\{0, -x_i\}$, $s = h - Gx$. Thus $x_i^+ - x_i^- = x_i$ and $x^+ - x^- = x$. It is easily verified that x^+, x^-, s are feasible in the standard form LP, with objective value $c^T x^+ - c^T x^- + d$. Hence, for each feasible point in the original LP we can find a feasible point in the standard form LP with the same objective value. If all points feasible in the original LP have their equivalent in the standard form LP, it might be that some point feasible in the standard form LP is NOT feasible in the original LP AND has a lower objective value in the standard form LP. This implies that the optimal value of the standard form LP is less than or equal to the optimal value of the original problem.

Conversely, suppose x^+, x^-, s are feasible in the standard form LP. Define $x = x^+ - x^-$. It is clear that x is feasible for the original LP, with objective value $c^T x + d = c^T x^+ - c^T x^- + d$. Hence, for each feasible point in the standard form LP we can find a feasible point in the original LP with the same objective value. This implies that the optimal value of the standard form LP is greater than or equal to the optimal value of the original. We conclude that the optimal values are equal.